

All Arduino Codes

1. Arduino LED Blink

```
void setup() {  
    pinMode(10, OUTPUT); // Set pin 13 as output  
}  
  
void loop() {  
    digitalWrite(10, HIGH); // Turn LED on  
    delay(500);           // Wait for 1 second  
    digitalWrite(10, LOW); // Turn LED off  
    delay(500);           // Wait for 1 second  
}
```

2. Arduino Switch Interfacing

```
void setup() {  
    // Set digital pins 4 to 7 as inputs  
    pinMode(4, INPUT);  
    pinMode(5, INPUT);  
    pinMode(6, INPUT);  
    pinMode(7, INPUT);  
  
    // Set digital pins 8 and 9 as outputs  
    pinMode(8, OUTPUT); // RC1
```

```

pinMode(9, OUTPUT); // RC2

// Turn off both outputs initially
digitalWrite(8, LOW);
digitalWrite(9, LOW);

}

void loop() {
    // Read the inputs (RB4 to RB7)
    int rb4 = digitalRead(4);
    int rb5 = digitalRead(5);
    int rb6 = digitalRead(6);
    int rb7 = digitalRead(7);

    // Create a byte value from RB4-RB7
    byte val = (rb4 << 4) | (rb5 << 5) | (rb6 << 6) | (rb7 << 7);

    // Compare input pattern and control output
    if (val == 0xE0) {
        // 1110 0000 → RB5, RB6, RB7 HIGH; RB4 LOW
        digitalWrite(8, HIGH);
        digitalWrite(9, LOW);
    }
    else if (val == 0xD0) {
        // 1101 0000 → RB4, RB6, RB7 HIGH; RB5 LOW
        digitalWrite(8, LOW);
        digitalWrite(9, HIGH);
    }
}

```

```
}  
  
else if (val == 0xB0) {  
    // 1011 0000 → RB4, RB5, RB7 HIGH; RB6 LOW  
    digitalWrite(8, HIGH);  
    digitalWrite(9, HIGH);  
}  
  
else if (val == 0x70) {  
    // 0111 0000 → RB4, RB5, RB6 HIGH; RB7 LOW  
    digitalWrite(8, LOW);  
    digitalWrite(9, LOW);  
}  
  
else {  
    // Default case: turn off both outputs  
    digitalWrite(8, LOW);  
    digitalWrite(9, LOW);  
}  
  
delay(50); // Short delay for stability  
}
```

3. Arduino LCD

```
#include <LiquidCrystal.h>

// initialize the library with the numbers of the interface pins
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

void setup() {
  lcd.begin(16, 2);      // set up the LCD's number of columns and rows
  lcd.print("ARDUINO LCD"); // Print a message to the LCD
}

void loop() {
  // nothing to do here
}
```

4. Arduino ADC-PWM

```
#define Analog_in PC1    // A1 is the analog input pin

#define Analog_Out PD6    // Digital pin 3 (PWM output)


void setup() {

    pinMode(Analog_in, INPUT);

    pinMode(Analog_Out, OUTPUT);

}


void loop() {

    int value = analogRead(Analog_in);    // Read analog value (0–1023)

    int pwmValue = map(value, 0, 1023, 0, 255); // Convert to 8-bit PWM

    analogWrite(Analog_Out, pwmValue);    // Write PWM output

}
```

5. Arduino UART

UART Master Code

```
#define PIN_1 13

#define PIN_2 12

#define PIN_3 11

#define PIN_4 10


void setup() {

    Serial.begin(9600);
```

```
pinMode(PIN_1, INPUT_PULLUP);  
pinMode(PIN_2, INPUT_PULLUP);  
pinMode(PIN_3, INPUT_PULLUP);  
pinMode(PIN_4, INPUT_PULLUP);  
delay(1000);  
}  
  
void loop() {  
  if (digitalRead(PIN_1) == LOW) {  
    Serial.println("A");  
  } else if (digitalRead(PIN_2) == LOW) {  
    Serial.println("B");  
  } else if (digitalRead(PIN_3) == LOW) {  
    Serial.println("C");  
  } else if (digitalRead(PIN_4) == LOW) {  
    Serial.println("D");  
  }  
  
  delay(200); // Avoid flooding the serial port  
}
```

UART Slave Code

```
#define LED1 7

#define LED2 6

void setup() {

    Serial.begin(9600);

    pinMode(LED1, OUTPUT);

    pinMode(LED2, OUTPUT);

    delay(1000);

}

void loop() {

    if (Serial.available() > 0) {

        String data = Serial.readStringUntil('\n');

        data.trim(); // Remove any extra whitespace or \r

        if (data == "A") {

            digitalWrite(LED1, HIGH);

            digitalWrite(LED2, LOW);

        } else if (data == "B") {

            digitalWrite(LED1, LOW);

            digitalWrite(LED2, HIGH);

        } else if (data == "C") {

            digitalWrite(LED1, HIGH);

            digitalWrite(LED2, HIGH);

        } else {

            digitalWrite(LED1, LOW);

            digitalWrite(LED2, LOW);

        }

    }

}
```

}

}

}